

Madoka Masui
4-7-14 Zaimokuza, Kamakura, 248-0013 Japan
lucaremm@gmail.com

February 27, 2026

Dear Editor,

I am pleased to submit the manuscript entitled “*Irreversibility-Constrained Governance: Preserving Human Responsibility under Finite Oversight Capacity*” for consideration in *Ethics and Information Technology*.

The manuscript addresses a structural problem emerging in AI-mediated institutional environments: under conditions of continuous updating and adaptive optimization, human responsibility may erode not through misattribution, but through the progressive depletion of oversight capacity prior to irreversible outcomes.

While existing governance frameworks such as Meaningful Human Control clarify alignment and traceability conditions, they do not model the dynamic evolution of institutional oversight capacity itself. The proposed framework, Irreversibility-Constrained Governance (ICG), introduces a capacity-based model of institutional irreversibility and formalizes a velocity constraint regulating the rate at which systems approach configurations in which evaluative and corrective judgment become structurally ineffective.

The contribution of the manuscript is threefold:

1. It distinguishes biological irreversibility from institutional irreversibility as analytically separate but structurally related phenomena.
2. It models oversight capacity as a dynamic variable subject to asymmetric depletion and recovery, incorporating institutional hysteresis.
3. It proposes structured freeze mechanisms as recovery operations that preserve the evaluative space necessary for responsible judgment.

The framework is instantiated in a high-risk ICU sepsis decision-support context to demonstrate that the structural dynamics of capacity depletion and recovery are most consequential where biological irreversibility represents an absolute downstream threshold.

The manuscript does not argue against technological optimization. Rather, it proposes that governance must regulate the trajectory of system evolution under conditions of finite human capacity. Its objective is to preserve the structural conditions under which human responsibility remains operational rather than merely formal.

This manuscript has not been published elsewhere and is not under consideration by any other journal. There are no conflicts of interest to declare. This work received no external funding and was conducted without institutional affiliation.

Thank you for your consideration.

Sincerely,

Madoka Masui
Independent Researcher
4-7-14 Zaimokuza, Kamakura, 248-0013 Japan
lucaremm@gmail.com